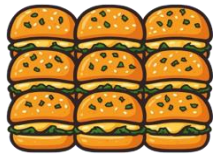
Stanford CS224v Course
Agentic AI

Lecture 10

Systematic Review of Long Document Sets



SLIDERS: Evidence Synthesis with Autonomous Database Reconciliation for Systematic Reviews

Harshit Joshi Monica Lam

Lecture Goal

- **What are systematic reviews and why are they hard?**
  - Failures of LLMs, and chunking methods
- **SLIDERS**
  - Key concept: Reconciliation
  - Building autonomous database reconciliation agent
  - The lack of benchmarks and what it takes for testing SOTA methods

Systematic Reviews

- Cornerstone of **evidence-based research** across medicine, social sciences, policy analysis, finance, law, and other technical domains

REVIEW

Systematic review of conservation interventions to promote voluntary behavior change

Laura Thomas-Walters[1] | Jamie McCallum[2] | Ryan Montgomery[3]
Anita K. Y. Wan[5] | Diogo Veríssimo[4]

Critical Criminology (2024) 32:761–801
https://doi.org/10.1007/s10612-024-09768-2

Indigent Injustice: A Systematic Review and Meta-analysis of People's Criminal Legal Outcomes

RESEARCH

OPEN ACCESS

Check for updates

Ultra-processed food exposure and adverse health outcomes: umbrella review of epidemiological meta-analyses

Melissa M Lane,[1] Elizabeth Gamage,[1] Shutong Du,[2,3] Deborah N Ashtree,[1]
nelia J McGuinness,[1] Sarah Gauci,[1,4] Phillip Baker,[5] Mark Lawrence,[6] Casey M Rebholz,[2,3]
rnard Srour,[7] Mathilde Touvier,[7] Felice N Jacka,[1,8,9] Adrienne O'Neil,[1] Toby Segasby,[10]
olfgang Marx[1]

STRACT
JECTIVE
evaluate the existing meta-analytic evidence of
sociations between exposure to ultra-processed
Nova food classification
alth outcomes.

iew of existing meta-analyses.

base, and the Cochrane
Reviews, as well as manual
sts from 2009 to June 2023.

OR SELECTING STUDIES
meta-analyses of cohort,
ss sectional study designs.
ity of evidence, pre-specified
criteria were applied,
class I"), highly suggestive
"class III"), weak ("class
IV"), or no evidence ("class V"). The quality of
evidence was assessed using the GRADE (Grading

supported direct associations between greater ultra-processed food exposure and higher risks of incident cardiovascular disease related mortality (risk ratio 1.50, 95% confidence interval 1.37 to 1.63; GRADE=very low) and type 2 diabetes (dose-response risk ratio 1.12, 1.11 to 1.13; moderate), as well as higher risks of prevalent anxiety outcomes (odds ratio 1.48, 1.37 to 1.59; low) and combined common mental disorder outcomes (odds ratio 1.53, 1.43 to 1.63; low). Highly suggestive (class II) evidence indicated that greater exposure to ultra-processed foods was directly associated with higher risks of incident all cause mortality (risk ratio 1.21, 1.15 to 1.27; low), heart disease related mortality (hazard ratio 1.66, 1.51 to 1.84; low), type 2 diabetes (odds ratio 1.40, 1.23 to 1.59; very low), and depressive outcomes (hazard ratio 1.22, 1.16 to 1.28; low), together with higher risks of prevalent adverse sleep related outcomes (odds ratio 1.41, 1.24 to 1.61; low), wheezing (risk ratio 1.40, 1.27 to 1.55; low), and obesity (odds ratio 1.55, 1.36 to 1.77; low). Of the remaining 34 pooled analyses, 21 were graded as suggestive or weak strength (class III-IV)

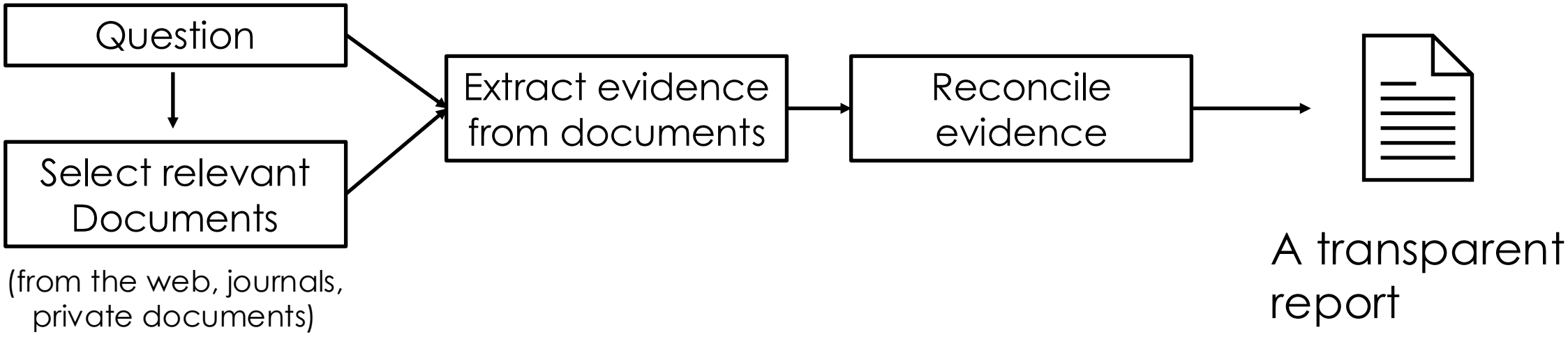
Meta-analysis of field experiments shows no change in racial discrimination in hiring over time

Lincoln Quillian[a,b,1], Devah Pager[c,d], Ole Hexel[a,e], and Arnfinn H. Midtbøen[f]

[a]Department of Sociology, Northwestern University, Evanston, IL 60208; [b]Institute for Policy Research, Northwestern University, Evanston IL 60208; [c]Department of Sociology, Harvard University, Cambridge, MA 02138; [d]Kennedy School of Government, Harvard University, Cambridge MA 02138; [e]Sciences Po, Observatoire Sociologique du Changement (OSC), CNRS, 75007 Paris, France; and [f]Institute for Social Research, N-0208 Oslo, Norway

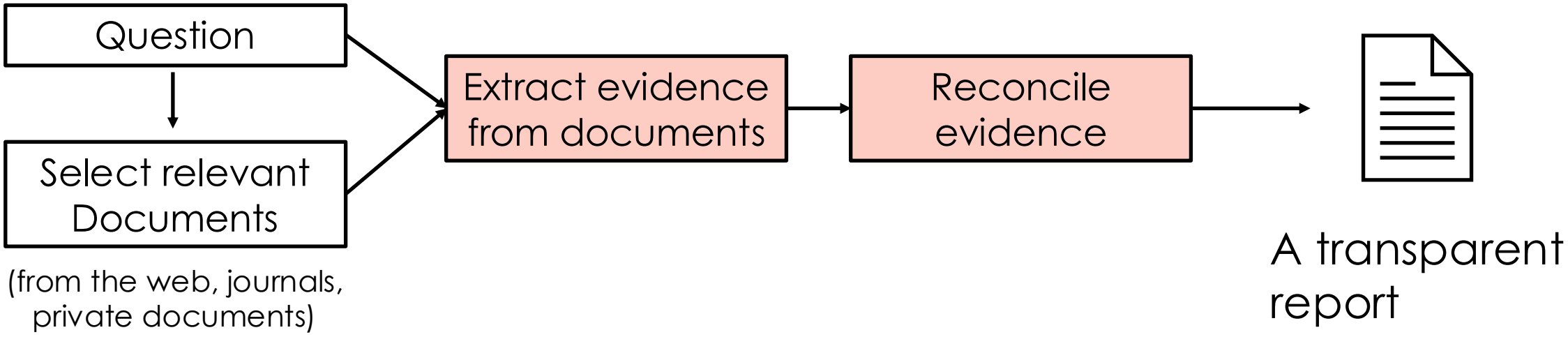
Edited by Douglas S. Massey, Princeton University, Princeton, NJ, and approved August 8, 2017 (received for review April 14, 2017)

Usual systematic review process



Takes 67.3 weeks to complete one review

Usual systematic review process



Number of documents ranges from 20 to 72 documents per review!

Types of questions over document set

Document Set	Question	Type
Collect of Restaurant Reviews	What are some good restaurants in Palo Alto?	Simple Lookup
Financial Document	How many AI companies have positive net profit?	Count all evidences
Clinical Documents	What patient and disease factors should be used to select patients with stage I–III hormone receptor-positive for neoadjuvant therapy?	Compare all evidences
Historical Documents	Which communities were flourishing under Chandragupta Murya empire and what led to their demise?	Tracing a story with ordering

Why is systematic review difficult?

Long Context	Large set of long documents
Retrieve All Instances	Need to find every relevant information – high recall
Reconcile	Compare all the evidences
Transparent Reporting	Require context about every evidence to make a coherent story

Systematic reviews for oncology

1. Identify eligible documents Sheldon
2. Extract characteristics of clinical studies SLIDERS
3. Extract or compute significance SLIDERS
4. Perform literature review based on underlying documents. SLIDERS
5. Curate the final set of guidelines SLIDERS

Why is systematic review difficult?

What patient and disease factors should be used to select patients with stage I-III hormone receptor-positive, HER2-negative breast cancer to whom neoadjuvant therapy should be offered?

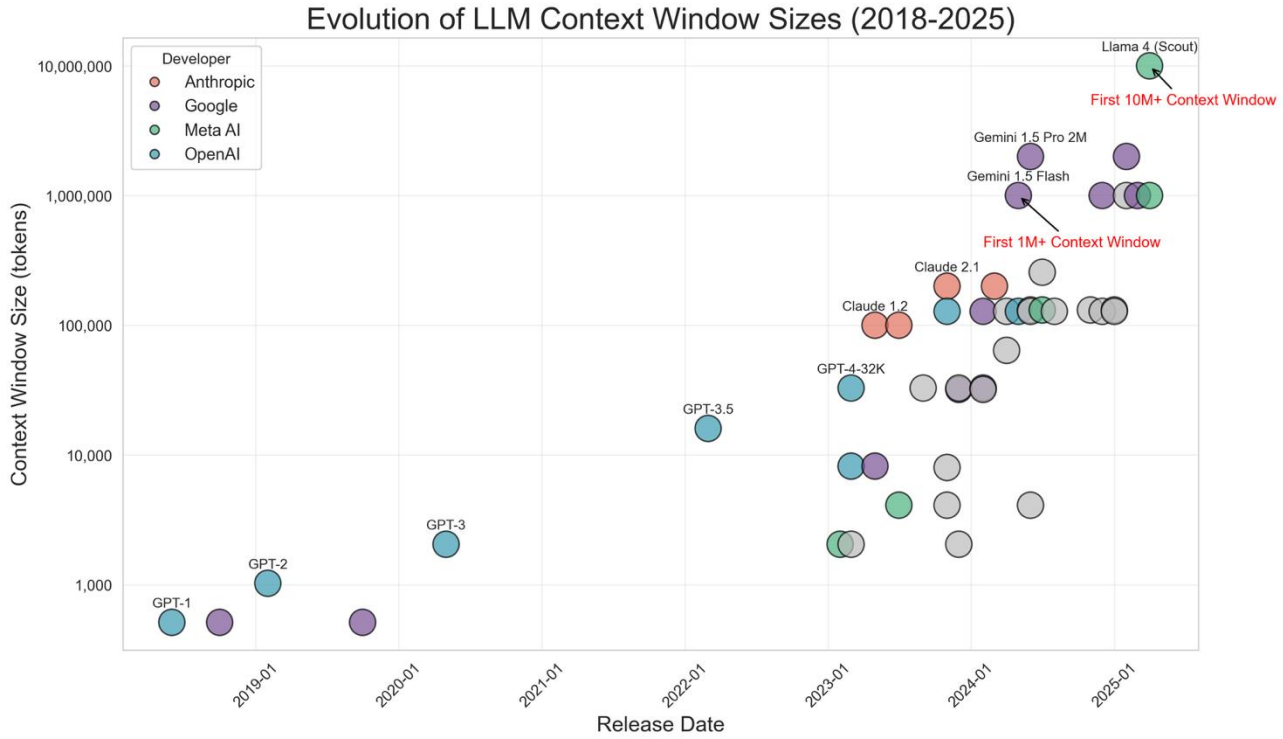
Long Context	*65 documents*
Retrieve All Instances	*A single document contains several evidences.* *Example: need to get underlying number to compute significance*
Reconcile	*Need to check for which value to keep.* *Example total patients admitted vs total patients in actual trial*
Transparent Reporting	*Build the report based on additional context, talk about caveats and contextualize the information*

Outline

- **Can LLMs perform systematic reviews?**
- High-level approaches: Training vs. Chunking
- Introduction to SLIDERS
- Design
- Evaluation

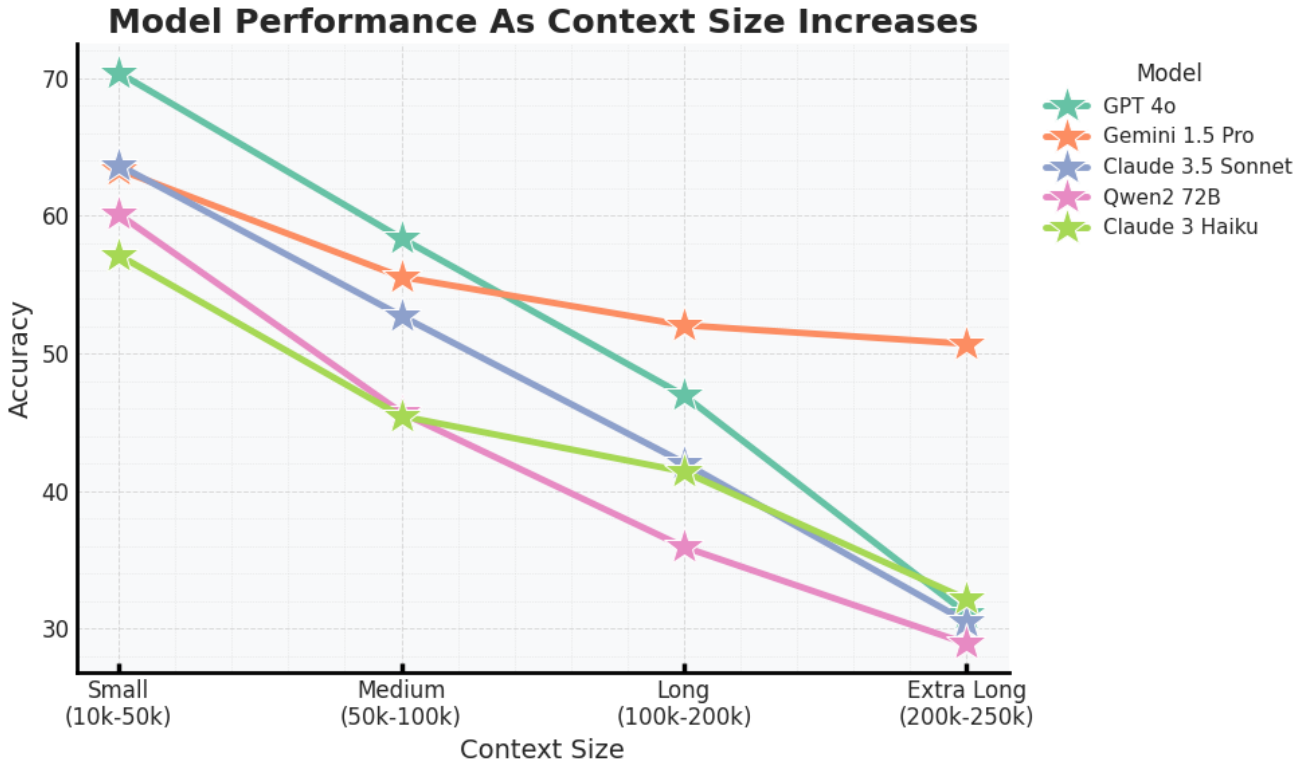
LLMs Long-Context Capability

- Ever increasing context window of LLMs
- Near perfect needle in a haystack capabilities

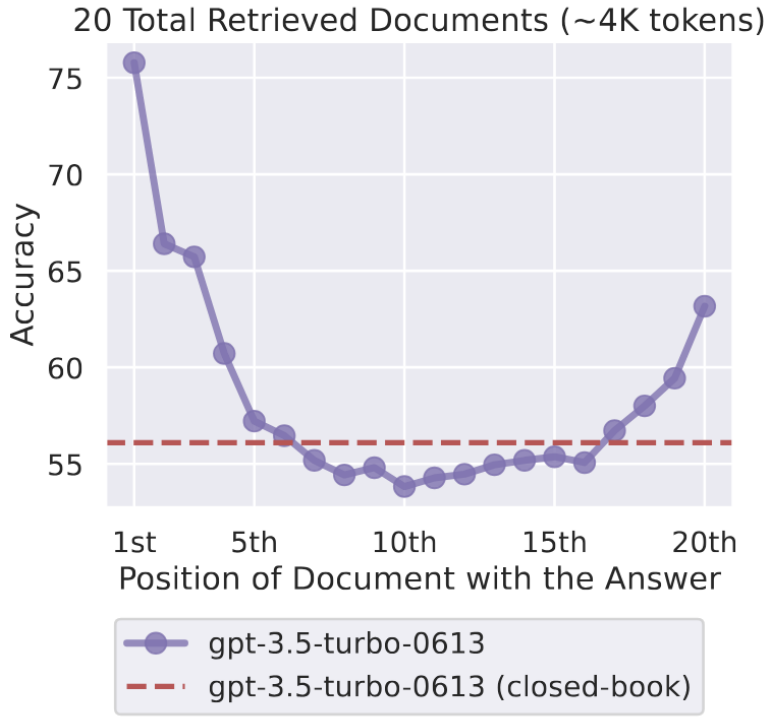


LLMs Lack Precision for Long Contexts

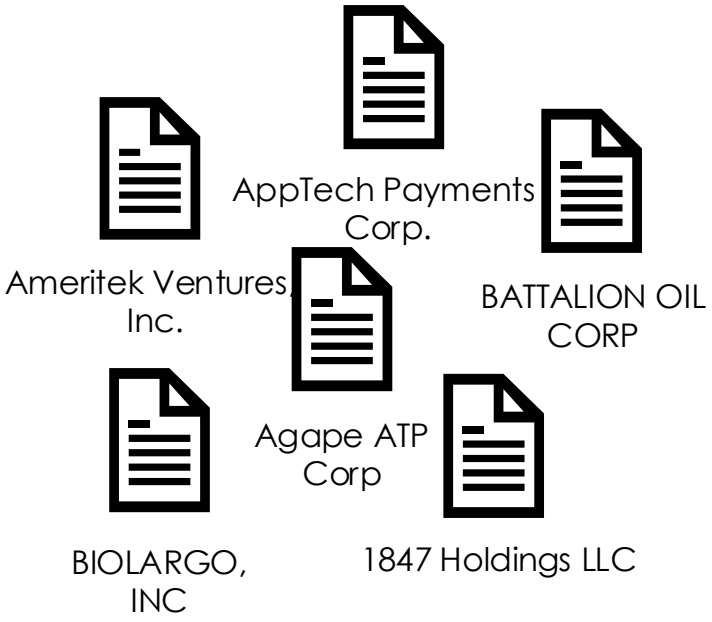
- Context length increases → performance for the same task decreases
- Attention is not uniform across the entire context window.



"Loong Benchmark" Wang et. al 2024



"Lost in the middle" Liu et. al 2023



Examples of Imprecision

What is the accounts payable of BIOLARGO, INC.?

GPT 4.1:

The accounts payable of BIOLARGO, INC. as of March 31, 2024, is $1,740,000.

Line 271: accounts payable and accrued expenses
Line 1735: accounts payable,
separate from accrued expenses (line 1737)

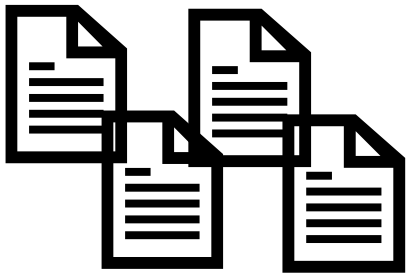
	Liabilities and stockholders' equity		
	Current liabilities:		
271	Accounts payable and accrued expenses	$ 1,740	$ 1,488
272	Clyra Medical accounts payable and accrued expenses	772	397
273	Clyra Medical debt obligations	234	234
274	Debt obligation	66	66
275	Contract liabilities	261	303
276	Lease liability		
277	Deposits		
278	Total current liabil		
279			

GPT 4.1 Answer

	Category	BioLargo	ONM	BLEST	Canada	BETI	BEST	Intercompany	Totals
1731									
1732									
1733								amounts	
1734									
1735	Accounts payable	$ 200	$ 1,357	$ 62	$ 44	$ 24	$ –	$ (24)	$ 1,663
1736									
1737	Accrued payroll	13	39	22	–	–	3	–	77
1738									
1739	Total							$	1,740
1740									
1741									
1742									
1743	As of December 31, 2023, accounts payable and accrued expenses included the following (in thousands):								

Correct Answer

Examples of Imprecision



Given 4 academic papers

We hope you will carefully study the provided papers and determine the citation relationships between them.

1. Reference: references are about what the given paper is using.

2. Citation: citations are about who is using the given paper.

The paper you need to analyze:
Risk Taxonomy, Mitigation, and Assessment Benchmarks of Large Language Model Systems

GPT

{
"Reference": [],
'Citation':
['# Unbridled Icarus: A Survey of the Potential Perils of Image Inputs in Multimodal Large Language Model Security ']
}

Gold Answer

{
'Reference': ['# Evil Geniuses: Delving into the Safety of LLM-based Agents ',
'# Towards Optimal Statistical Watermarking '],
'Citation':
['# Unbridled Icarus: A Survey of the Potential Perils of Image Inputs in Multimodal Large Language Model Security ']
}

Misses several references

Contexts are never long enough

- Is 1 million tokens enough?
  - A pubmed article is around 100k tokens.
    - Can fit around 10 pubmed papers can fit in
  - A financial document (10-Q) is around 300k tokens.
    - Can fit around 3 financial document
  - A book 300 page book is about 2M tokens
    - Cannot even fit a single book in context!

Outline

- Can LLMs perform systematic reviews?
- **High-level approaches: Training vs. Chunking**
- Introduction to SLIDERS
- Design
- Evaluation

Approaches

- Training based: to improve precision
- Chunking based: to improve precision and scaling
  - Representations for chunks
    - Natural language
    - Structured representation
- Coding Agents

Training Based

QwenLong-L1: Towards Long-Context Large Reasoning Models with Reinforcement Learning

Fanqi Wan, Weizhou Shen, Shengyi Liao, Yingcheng Shi, Chenliang Li,
Ziyi Yang, Ji Zhang, Fei Huang, Jingren Zhou, Ming Yan*
Tongyi Lab, Alibaba Group

Self-Taught Agentic Long-Context Understanding

Yufan Zhuang[1,2], Xiaodong Yu[1], Jialian Wu[1], Ximeng Sun[1], Ze Wang[1],
Jiang Liu[1], Yusheng Su[1], Jingbo Shang[2], Zicheng Liu[1], Emad Barsoum[1]
[1]AMD, [2]UC San Diego

Cartridges: Lightweight and general-purpose long context representations via self-study

Sabri Eyuboglu [1*] Ryan Ehrlich [1*] Simran Arora [1,2*] Neel Guha [1] Dylan Zinsley [3] Emily Liu [1]
Will Tennien [1] Atri Rudra [3] James Zou [1] Azalia Mirhoseini [1] Christopher Ré [1]
[1]Stanford University [2] Caltech [3]University at Buffalo * Equal contribution

ALR²: A Retrieve-then-Reason Framework for Long-context Question Answering

Huayang Li[♢,♡,*] Pat Verga[♡] Priyanka Sen[♡] Bowen Yang[♡] Vijay Viswanathan[♣,♡]
Patrick Lewis[♡] Taro Watanabe[♢] Yixuan Su[♡]
♡ Cohere ♢Nara Institute of Science and Technology ♣ Carnegie Mellon University

MDCure: A Scalable Pipeline for Multi-Document Instruction-Following

Gabrielle Kaili-May Liu[1] Bowen Shi[1] Avi Caciularu[2]
Idan Szpektor[2] Arman Cohan[1]

[1]Yale University [2]Google Research

Large Language Models Can Self-Improve in Long-context Reasoning

Siheng Li[♡] Cheng Yang[♡] Zesen Cheng[♠] Lemao Liu[♢] Mo Yu[♢]
Yujiu Yang[♣] Wai Lam[♡]
♡The Chinese University of Hong Kong
♠Peking University ♣Tsinghua University ♢Tencent

Never Lost in the Middle: Mastering Long-Context Question Answering with Position-Agnostic Decompositional Training

Junqing He, Kunhao Pan, Xiaoqun Dong,
Zhuoyang Song, Yibo Liu, Qianguo Sun,
Yuxin Liang, Hao Wang, Enming Zhang, Jiaxing Zhang
International Digital Economy Academy, Shenzhen, China
hejunqing@idea.edu.cn

Training-Based Methods

QwenLong-L1:

- Curate a long-context dataset using existing benchmarks,
- Progressively increase the context size with RL
  using rule-based verification and LLM-as-a-judge
- Improve over baseline models,
  but not SOTA with proprietary/open-source models

Cartridges

- Train KV cache on a document set by generating & answering questions
- Reuse cache to answer new questions across the trained document set
- Lose to complete in-context documents

"QWENLONG-L1: Towards Long-Context Large Reasoning Models with Reinforcement Learning" Wan et. al 2025
"Cartridges: Lightweight and general-purpose long context representations via self-study", Eyuboglu et. al 2025

Is training all we need?

What training can improve?

- Use of information within a model's context
- Long context reasoning on trained tasks

What does not guarantee?

- Coverage of a collection large than the context window
- Exact enumeration and consistent outputs across inferences
- Interface for interpretability

Training is complementary -- not by itself a solution for exhaustive evidence synthesis.

Which problems can training solve?

Long Context	Training already fails here
Retrieve All Instances	
Reconcile	
Transparent Reporting	

Chunking-Based

RAG Methods

LongRAG: A Dual-Perspective Retrieval-Augmented Generation Paradigm for Long-Context Question Answering

Qingfei Zhao[1,2,†], **Ruobing Wang**[1,2], **Yukuo Cen**[4], **Daren Zha**[1], **Shicheng Tan**[3], **Yuxiao Dong**[3], **Jie Tang**[3,*]
[1]Institute of Information Engineering, Chinese Academy of Sciences;
[2]School of Cyber Security, University of Chinese Academy of Sciences;
[3]Tsinghua University; [4]Zhipu AI
{zhaoqingfei, wangruobing, zhadaren}@iie.ac.cn, yukuo.cen@zhipuai.cn
tsctan@foxmail.com, {yuxiaod, jietang}@tsinghua.edu.cn

HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models

Bernal Jiménez Gutiérrez
The Ohio State University

Yiheng Shu
The Ohio State University

Yu Gu
The Ohio State University

Michihiro Yasunaga
Stanford University

Yu Su
The Ohio State University

From Local to Global: A GraphRAG Approach to Query-Focused Summarization

Darren Edge[1†] **Ha Trinh**[1†] **Newman Cheng**[2] **Joshua Bradley**[2] **Alex Chao**[3]

Apurva Mody[3] **Steven Truitt**[2] **Dasha Metropolitansky**[1] **Robert Osazuwa Ness**[1]

Jonathan Larson[1]

[1]Microsoft Research
[2]Microsoft Strategic Missions and Technologies
[3]Microsoft Office of the CTO

{daedge,trinhha,newmancheng,joshbradley,achao,moapurva,
steventruitt,dasham,robertness,jolarso}@microsoft.com

[†]These authors contributed equally to this work

RAG Methods

- Technical data is usually spread across different paragraph and documents

Debt - Additional Information (Details)$ in Thousands		3 Months Ended
Sep. 09, 2022USD ($)	Nov. 30, 2025USD ($)	Nov. 30, 2024USD ($)
Debt Instrument [Line Items]		
Borrowings on Revolving Credit		$ 0
Senior Credit Facilty	$ 600,000	
Long-term debt, net		179,710
Long-term Debt		188,750
Senior Credit Facility - Term Loan		

CONDENSED CONSOLIDATED BALANCE SHEETS - USD () in Thousands	Nov. 30, 2025	Aug. 31, 2025
Current assets		
Cash and cash equivalents	$ 139,047	$ 151,558
Current maturities of long-term debt	41,698	42,944
Trade accounts payable	21,746	28,108
Long-term Debt, Current Maturities	8,750	7,500
Income taxes payable	9,841	5,425

Note 6. DebtThe following is a summary of the Company's long-term indebtedness (in thousands):November 30, 2025August 31, 2025Senior Credit FacilityRevolver$—$—Term Loan188,750190,000Total Senior Indebtedness188,750190,000Less: Current maturities of long-term debt(8,750)(7,500)Debt issuance costs(290)(332)Total long-term debt, less current maturities$179,710$182,168Senior Credit FacilityOn September 9,

suramentsThe Company assesses the inputs used to measure the fair value of financial sing a three-tier hierarchy. Level 1 inputs include unadjusted quoted prices for identical e most observable. Level 2 inputs include quoted prices for similar assets and observable t rates, foreign currency exchange rates, commodity rates and yield curves. Level 3 inputs the market and include management's own judgments about the assumptions market e in pricing an asset or liability.The fair value of the Company's cash and cash equivalents, accounts payable and variable rate long-term debt approximated book value at both nd August 31, 2025 due to their short-term nature and the fact that the interest rates rates. Foreign currency exchange contracts and interest rate swaps are recorded at fair f the Company's foreign currency exchange contracts was a net liability of less than $0.1 t of less th 31, 2025, respectively.The fair 's interest y of $0.3 million and $4.6 0, 2025, re d a liability of $4.7 million at ectively, (the Company's interest rate ent hedge sed on quoted inactive market lassified a

The following is a summary of the Company's long-term indebtedness (in thousands):November 30, 2025August 31, 2025Senior Credit FacilityRevolver$—$—Term Loan188,750190,000Total Senior Indebtedness188,750190,000Less: Current maturities of long-term debt(8,750)(7,500)Debt issuance costs(290)(332)Total long-term debt, less current maturities$179,710$182,168

RAG Methods

- Technical data is usually spread across different paragraphs and documents
- Need to collect all the evidence
  -- not just a small set of semantically relevant chunks
- Increase the number of passages to retrieve again
  leads to yet another long-context problem.

Which problems can RAG solve?

Long Context	RAG can cover all documents
Retrieve All Instances	Fails miserably
Reconcile	
Transparent Reporting	

Coding Agents

- Increasingly people are using coding agents for document QA
- Agents with access to files execute shell commands to retrieve passages and generate answer
- Usually use grep/keyword search to find relevant passages

Can using grep or keyword search cover all the cases?

Chunking-based Methods

DocETL: Agentic Query Rewriting and Evaluation for Complex Document Processing

Shreya Shankar[1], Tristan Chambers[2], Tarak Shah[2], Aditya G. Parameswaran[1], Eugene Wu[3]
[1]UC Berkeley EECS, [2]BIDS Police Records Access Project, [3]Columbia University
{shreyashankar,tristan.chambers,tarak_shah,adityagp} @berkeley.edu, ewu@cs.columbia.edu

Chain of Agents: Large Language Models Collaborating on Long-Context Tasks

Yusen Zhang[♣*], Ruoxi Sun[◇], Yanfei Chen[◇], Tomas Pfister[◇], Rui Zhang[♣†], Sercan Ö. Arik[◇†]
[♣] Penn State University, [◇] Google Cloud AI Research
{yfz5488, rmz5227}@psu.edu, {ruoxis, yanfeichen, tpfister, soarik}@google.com

HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction

Bhaskarjit Sarmah
BlackRock
IN
bhaskarjit.sarmah@blackrock.com

Dhagash Mehta
BlackRock, Inc.
US
dhagashbmehta@gmail.com

Benika Hall
NVidia
US
bhall@nvidia.com

Rohan Rao
NVIDIA
US
rohrao@nvidia.com

Sunil Patel
NVidia
US
spatel@nvidia.com

Stefano Pasquali
BlackRock, Inc.
US
stefano.pasquali@blackrock.com

Minions: Cost-efficient Collaboration Between On-device and Cloud Language Models

Avanika Narayan[*1], Dan Biderman[*1,2,3], Sabri Eyuboglu[*1], Avner May[5], Scott Linderman[2,3], James Zou[4], Christopher Ré[1]

[1]Department of Computer Science, Stanford University
[2]Department of Statistics, Stanford University
[3]Wu Tsai Neurosciences Institute, Stanford University
[4]Departemnet of Biomedical Data Science, Stanford University
[5]Together AI
{avanikan,biderman,eyuboglu}@stanford.edu

GraphReader: Building Graph-based Agent to Enhance Long-Context Abilities of Large Language Models

Shilong Li[*1], Yancheng He[*1], Hangyu Guo[*1], Xingyuan Bu[*†‡1], Ge Bai[1], Jie Liu[2,3], Jiaheng Liu[1], Xingwei Qu[4], Yangguang Li[3], Wanli Ouyang[2,3], Wenbo Su[1], Bo Zheng[1]
[1]Alibaba Group [2]The Chinese University of Hong Kong
[3]Shanghai AI Laboratory [4]University of Manchester
zhuli.lsl@taobao.com, xingyuanbu@gmail.com

Long Context Scaling: Divide and Conquer via Multi-Agent Question-driven Collaboration

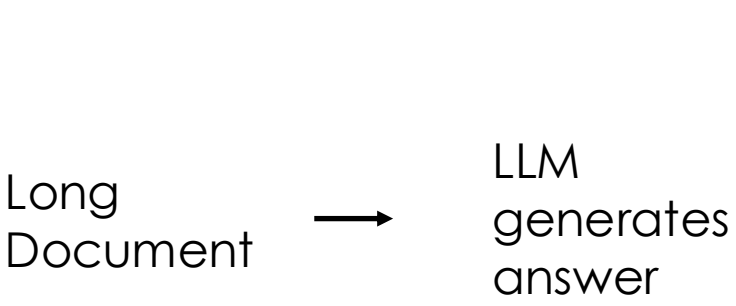
Sibo Xiao[1] Zixin Lin[1] Wenyang Gao[1 2] Hui Chen[1] Yue Zhang[2]

DataPuzzle: Breaking Free from the Hallucinated Promise of LLMs in Data Analysis

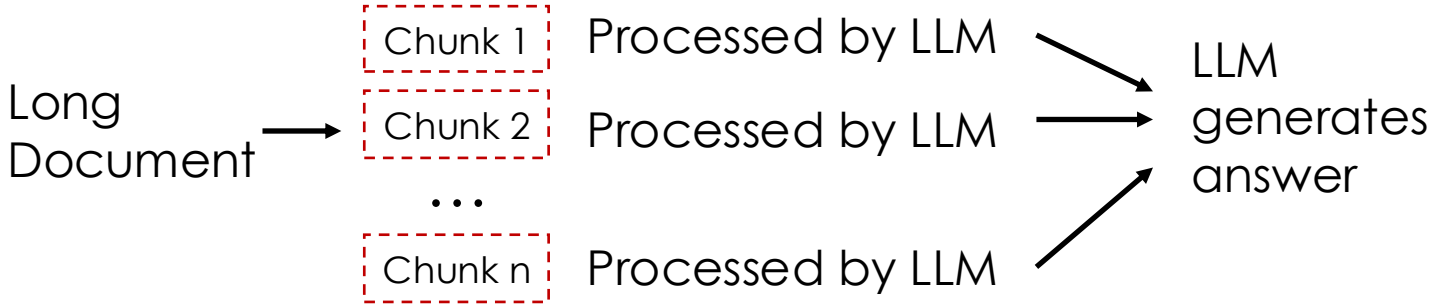
Zhengxuan Zhang, Zhuowen Liang Yin Wu, Teng Lin, Yuyu Luo, Nan Tang
The Hong Kong University of Science and Technology (Guangzhou)

Chunking-Based Methods

- Divide the context into multiple chunks.
- Answer question based on individual chunk
- Combine them together and get the final answer



Single step approach



Decomposing the problem

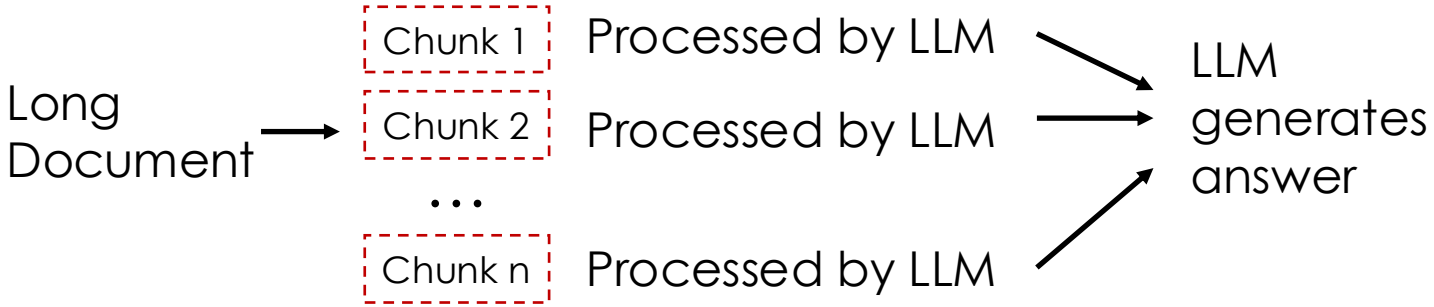
Chunking-Based Methods

- Chain of Agents:
  - An LLM processes a chunk and passes on summary w.r.t. the task to the next LLM call along with the next chunk.
  - Final agent synthesizes the answer based on last provided summary.
- DocETL:
  - User provides a schema for the representation
  - Each chunk is processed by an LLM that outputs structured data.
  - The user prompts the LLM to reduce/resolve the output from each chunk to answer the question.

"Chain of Agents: Large Language Models Collaborating on Long-Context Tasks" Zhang et. al 2024
"DocETL: Agentic Query Rewriting and Evaluation for Complex Document Processing" Shankar et. al 2024

Advantages of Chunking-Based Methods

- **Precision:**
  Processing each chunk individually increases attention to local details
- **Scalability:** Enables natural scalability to large document collections far beyond model limits



Decomposing the problem

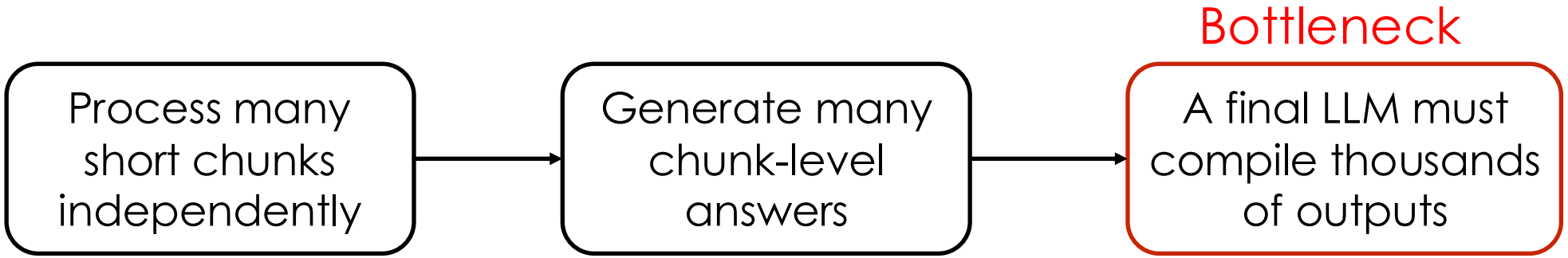
Disadvantages of Chunking

- Loss of context around each chunk
- Inconsistency in independent extractions
- Extracted evidence might be very “dirty”
- Collected evidence might still exceed context length of LLMs

Independently extracted information is “dirty”

- Extract similar information ➡ Results in overcounting
- Different extractions might have complimentary information ➡ Leads to missing information
- Contains conflicting information with different context ➡ Contradictory information and cannot form answer

Chunking solves input length, not answer synthesis



When synthesis is just "put everything in one more prompt," the long-context problem returns.

Is chunking all we need?

What chunking buys?

- Precision
- Scalability

Where chunking can fail?

- Inconsistent extraction
- Loss of context
- Dirty evidence collection
- Might still exceed the context

Which problems can chunking solve?

Long Context	Chunking methods work
Retrieve All Instances	Chunking methods can work
Reconcile	Loses context across chunks and fails
Transparent Reporting	Loses context across chunks and fails

For large document sets

Question	Type	Long context LLM	RAG	Chunking Methods
What are some good restaurants in Palo Alto?	Simple Lookup	✗	✓	✓
How many AI companies have positive net profit?	Count all evidences	✗	✗	✗
What patient and disease factors should be used to select patients with stage I–III hormone receptor-positive for neoadjuvant therapy?	Compare all evidences	✗	✗	✗
Which communities were flourishing under Chandragupta Murya empire and what lead to their demise?	Tracing a story with ordering	✗	✗	✗

Challenges of Chunking

1. **How do we represent the information in each chunk?**

2. **Problem answering separated into two steps**
   a. **Correct extraction of information chunk by chunk?**
      - Extraction Inconsistencies: might have different scale, units, etc
      - Loss of Context: interpretation might be different

   b. **Compile correct answer from the set of extracted information**
      - Extracted information is “dirty”
      - How to automatically find issues and fix them autonomously?

Our Solution: Chunk and Reconcile

1. **How do we represent the information in each chunk?**
   As rows in tables, database can scale
2. **Problem answering separated into two steps**
   a. **Correct extraction of information chunk by chunk?**
      - Extraction Inconsistencies: might have different scale, units, etc

      Automatically-induced schema with extraction rules
      - Loss of Context: interpretation might be different

      Contextualized information extraction
   b. **Compile correct answer from the set of extracted information**
      - Extracted information is "dirty"

      We discovered: duplications, incomplete information…
      - How to automatically find issues and fix them autonomously?

      Autonomous database reconciliation coding agent

Which problems can chunking-reconcile solve?

Long Context	Chunking methods work
Retrieve All Instances	Chunking methods can work
Reconcile	Need explicit reconciliation
Transparent Reporting	Need to preserve all the context

Kinds of Questions

SLIDERS

Question	Type	LLM	RAG	Chunking Methods	Chunk-Reconcile
What are some good restaurants in Palo Alto?	Simple Lookup	✗	✓	✓	✓
How many AI companies have positive net profit?	Count all evidences	✗	✗	✗	✓
What patient and disease factors should be used to select patients with stage I–III hormone receptor-positive for neoadjuvant therapy?	Compare all evidences	✗	✗	✗	✓
Which communities were flourishing under Chandragupta Murya empire and what lead to their demise?	Tracing a story with ordering	✗	✗	✗	✗

SLIDERS: Evidence Synthesis with Autonomous Database Reconciliation for Systematic Reviews*

Harshit Joshi **Priyank Shethia** **Jadelynn Dao** **Monica S. Lam**
Computer Science Department, Stanford University
{hj, lam}@cs.stanford.edu

To be presented at NeurIPS 2026, Sydney, Australia

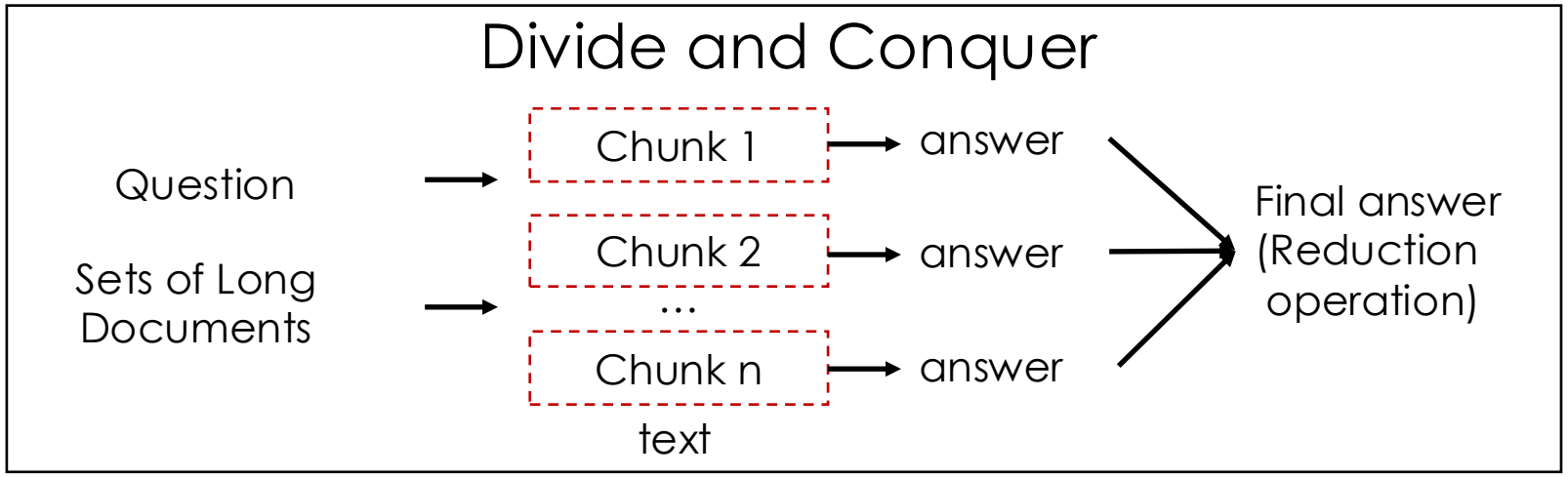
Outline

- Can LLMs perform systematic reviews?
- High-level approaches: Training vs. Chunking
- Introduction to SLIDERS
- **Design**
- Evaluation

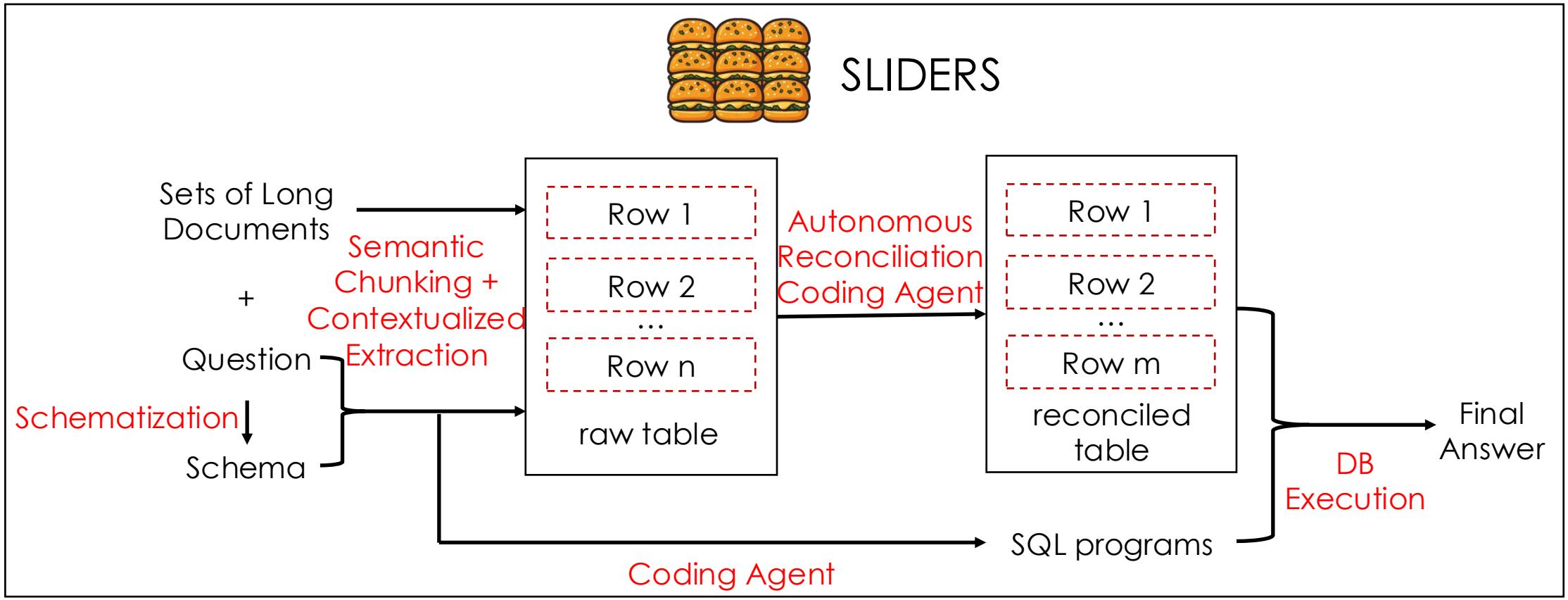
SLIDERS design

- **Database:** Use database for storing evidences
- **Schematization:** Automatically determines the database schema
- **Chunking and Extraction:** Extracts information chunk by chunk with provenance
- **Reconciliation:** Chunk-based extraction requires reconciliation
- **Autonomous database reconciliation coding agent:** Uses provenance and rationale
- **Coding agent:** To answer from reconciled tables

Divide and Conquer



- **Inconsistent extraction**
- **Loss of context**
- **Dirty evidence collection**
- **Might still exceed the context**



SLIDERS design

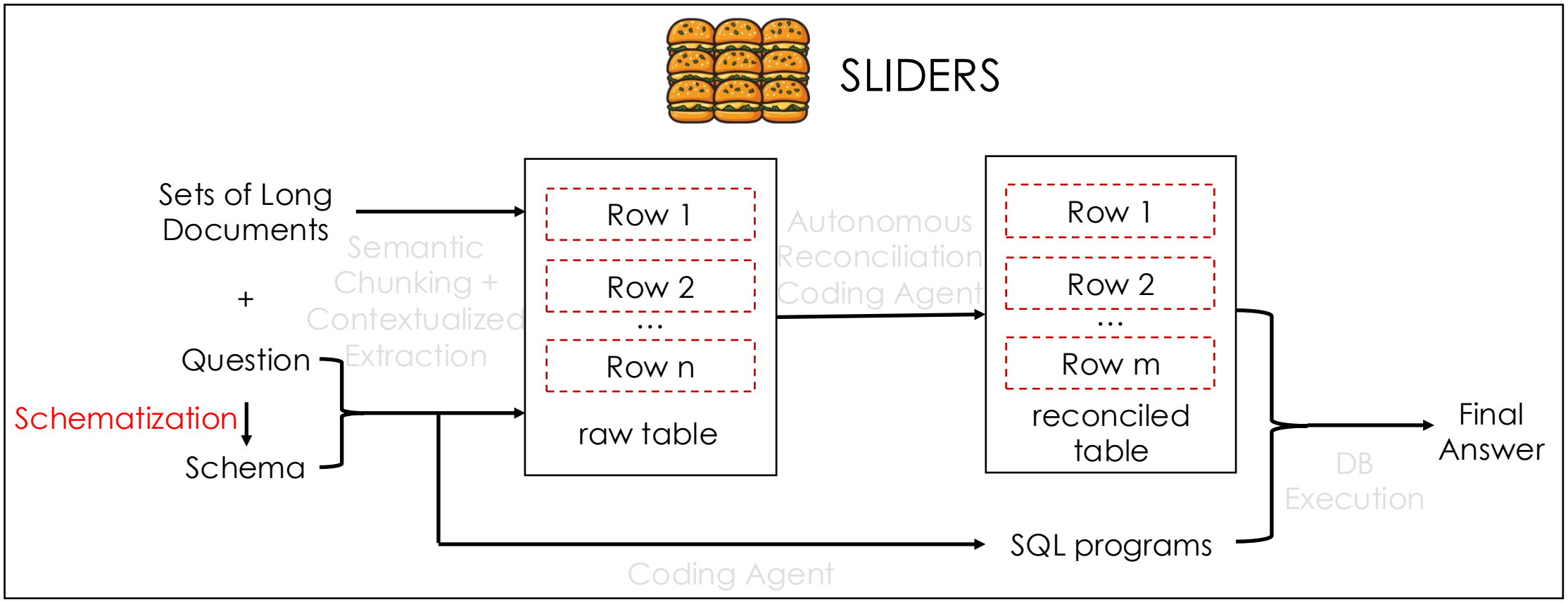
- **Database**
- Schematization
- Chunking and Extraction
- Reconciliation
- Autonomous database reconciliation coding agent
- Coding Agent for reconciled database

Advantages of using Database

- Can contain arbitrary amount of data
- Use columns to bring information together using groupby, projects etc
- LLMs are good at generating SQL

SLIDERS design

- Database
- **Schematization**
- Chunking and Extraction
- Reconciliation
- Autonomous database reconciliation coding agent
- Coding Agent for reconciled database


SLIDERS
Sets of Long Documents
+
Question
Schematization
Schema
Semantic Chunking + Contextualized Extraction
Row 1
Row 2
…
Row n
raw table
Autonomous Reconciliation Coding Agent
Row 1
Row 2
…
Row m
reconciled table
Final Answer
DB Execution
SQL programs
Coding Agent

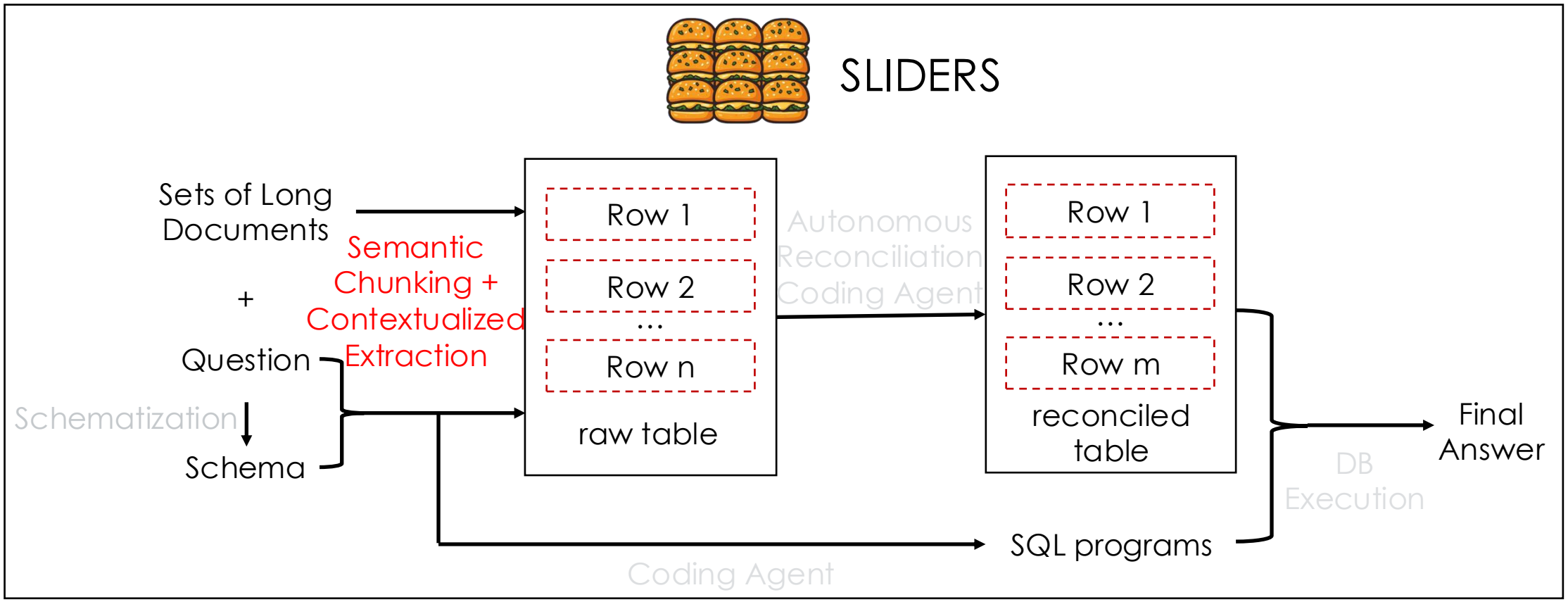
Schematization

- **Why?**
  - Users commonly refine their questions and will ask for followup questions
  - Can we guess the followup questions the user might ask?
- **How?** LLMs are capable of knowing what schema is required.
  - We need to provide sample documents
  - Explicitly prompt to find a broader latent schema and anticipate future questions
  - Need units and scale for extraction – happens in parallel.

Extracting extra information can **amortize cost** and don't have to read documents again for follow-up questions.

SLIDERS design

- Database
- Schematization
- **Chunking and Extraction**
- Reconciliation
- Autonomous database reconciliation coding agent
- Coding Agent for reconciled database

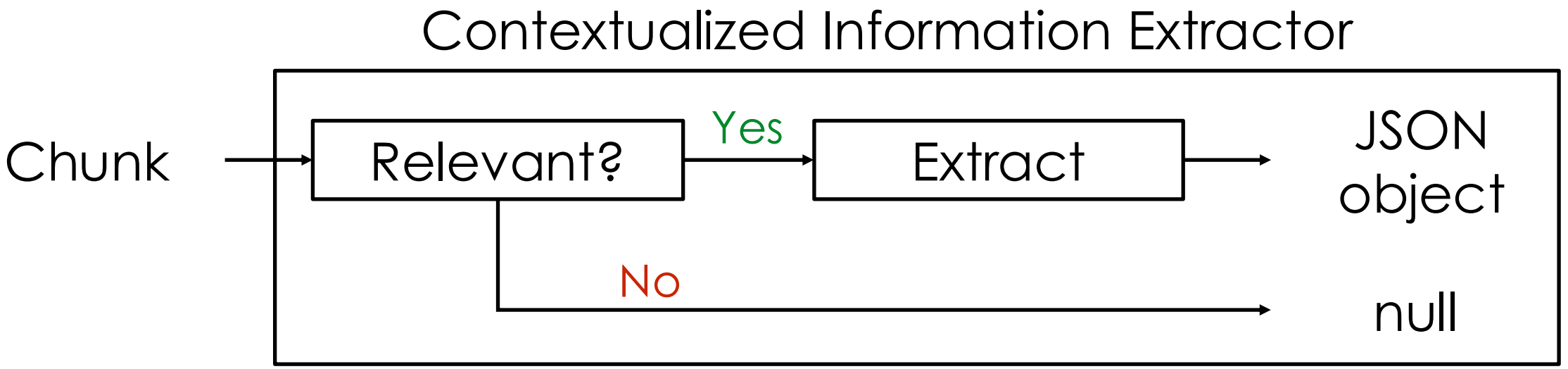

SLIDERS
Sets of Long Documents
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reconciled table
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Contextualized Extraction Technique

- LLMs still like to hallucinate
  - When given text contains no useful information (common for chunks)
  - When extracting a JSON object

→ Relevance gating.

- Use a lightweight LLM to decide if a chunk is relevant



SLIDERS design

- Database
- Schematization
- Chunking and Extraction
- **Reconciliation**
- Autonomous database reconciliation coding agent
- Coding Agent for reconciled database

Chunk-wise extraction can be “dirty”

How do long term borrowing patterns differ across companies and sectors?

row_id	page_number	document_name	company_name	period_end	long_term_debt
0	0	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	214.095
1	2	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	214.095
2	3	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	45.0
3	4	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	530.967
4	5	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO., INC.	2025-11-29	214.095
5	0	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	1368.738
6	2	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	1375.0
7	3	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	1368.738
8	4	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	

Extracted example from our 100 annual financial filings

Chunk-wise extraction can be “dirty”

row_id	page_number	document_name	company_name	period_end	long_term_debt
0	0	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	214.095
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8	4	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	

Chunk-wise extraction can lead to:

- Duplicates
- Incomplete information
- Conflicting information

Traditional database cleaning

- Past works focus on either detecting errors or fixing errors
- Traditionally use statistical methods and integrity constrains

  Example: two different cities connot have the same pincode

- Recent LLM methods identify typos, or find semantic outliers, or require labelled human data

  Example: datetime conversion, fix "united states of amerida" to "united states of America"

- Doesn't work when you need to address gross redundancies, inconsistencies and incompleteness caused by chunk-wise extraction

Need context for database cleaning

row_id	page_number	document_name	company_name	period_end	period_end_quote	period_end_rationale	long_term_debt	long_term_borrowings_quote	long_term_borrowings_rationale
0	0	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	['For the quarterly period ended November 29, 2025']	The period end date is explicitly stated as November 29, 2025, which is normalized to ISO date format YYYY-MM-DD.	214.095	['Long-term debt including obligations under finance leases | $ | 214,095']	The long-term borrowings are reported in the C obligations under finance leases' with a value o 214.095 millions USD.
1	2	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	['November 29, 2025']	The reporting period end date is explicitly stated multiple times in the notes and tables as November 29, 2025, normalized to ISO date format.	214.095	['Total long-term debt, including obligations under finance leases $ 214,095']	The total long-term debt including obligations u normalizes to 214.095 million USD.
2	3	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	['thirteen-week period ended November 29, 2025']	The period end date is explicitly stated multiple times as 'November 29, 2025'. Normalized to ISO date format 'YYYY-MM-DD'.	45.0	['We had net borrowings of $45.0 million on our credit facilities and private placement debt']	The text explicitly states 'net borrowings of $45 thirteen weeks ended November 29, 2025. This USD, so no further normalization needed.
3	4	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO INC	2025-11-29	['November 29, 2025']	The reporting period end date is explicitly stated multiple times in tables and text as November 29, 2025, normalized to ISO date format.	530.967	['Total debt $ 530,967']	The table under 'Liquidity and Capital Resourc which corresponds to $530.967 million USD lo financing arrangements.
4	5	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO., INC.	2025-11-29	['Nov. 29, 2025']	Period end date explicitly stated in the document header and financial statements.	214.095	['Total long-term debt, including obligations under finance leases $214,095,000']	Long-term borrowings amount explicitly stated 214.095 million USD.
5	0	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	['For the quarterly period ended November 30, 2025']	The period end date is explicitly stated in the document header and in the financial statements section as November 30, 2025, normalized to ISO date format.	1368.738	['Long-term debt | $ | 1,368,738 | November 30, 2025']	The Consolidated Balance Sheets section lists Converted from thousands USD to millions US
6	2	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	['as of November 30, 2025']	The reporting period end date is explicitly stated in the table header as November 30, 2025, normalized to ISO date format.	1375.0	['Total principal amount $ 1,375,000']	The total principal amount of long-term debt is million USD.
7	3	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	['as of November 30, 2025']	The period end date is explicitly stated multiple times in the document, including in the debt section and lease section, normalized to ISO date format YYYY-MM-DD.	1368.738	['Long-term debt(1) $ 1,368,738']	The long-term debt is explicitly stated in the de millions USD by dividing by 1000, resulting in 1
8	4	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC	2025-11-30	['November 30, 2025']	The reporting period end date is explicitly mentioned multiple times, including in tables and text, as November 30, 2025. Normalized to ISO date format.			Although the document references debt and le disclosure of long-term borrowings or long-ter

We can extract provenance, text span,
and extraction rationale

SLIDERS design

- Database
- Schematization
- Chunking and Extraction
- Reconciliation
- **Autonomous database reconciliation coding agent**
- Coding Agent for reconciled database

Database Agents

- Example SUQL
- User provides the problem → Agent generates the SQL
- Challenge: we need to identify the problem as well

SLIDERS: First database reconciliation agent that figures out the problem and comes up with the solution

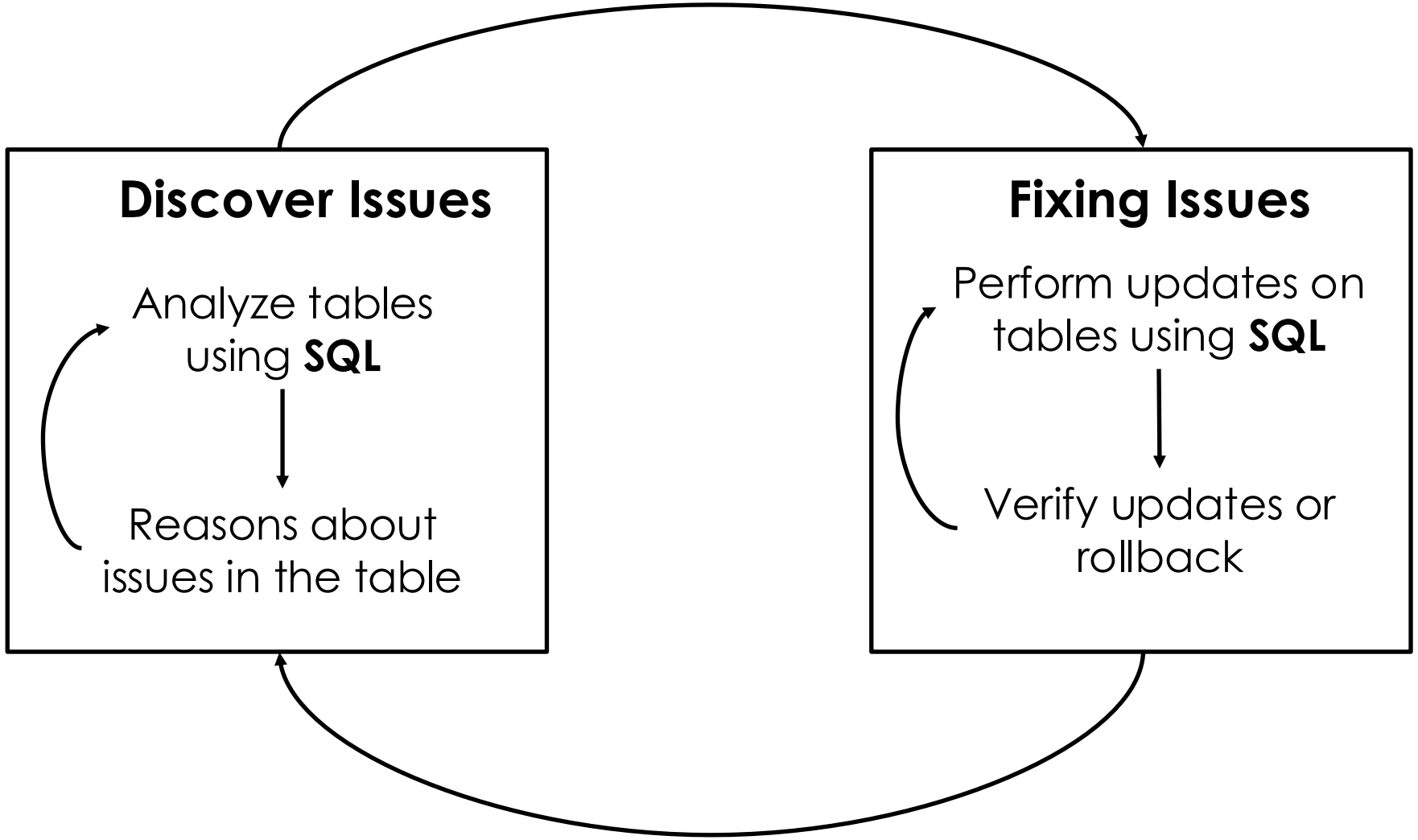
Can LLM agent do it?

- Will load the first few rows (10) and generalize the rule

  Example: The best value among the first 10 rows for long term debt is MAX. So for each company I will keep the maximum long term debt

- Will try to load the whole database!
  - Which exceeds the context if we have large amount of evidence
- Makes irreversible updates which are incorrect!

Autonomous Database Reconciliation Coding Agent



Autonomous Database Reconciliation Coding Agent

- Group the table by primary key
- Select primary key from the schema and inspecting the table
  - LLMs are inconsistent, use majority vote
- Provide explicit examples of primary keys

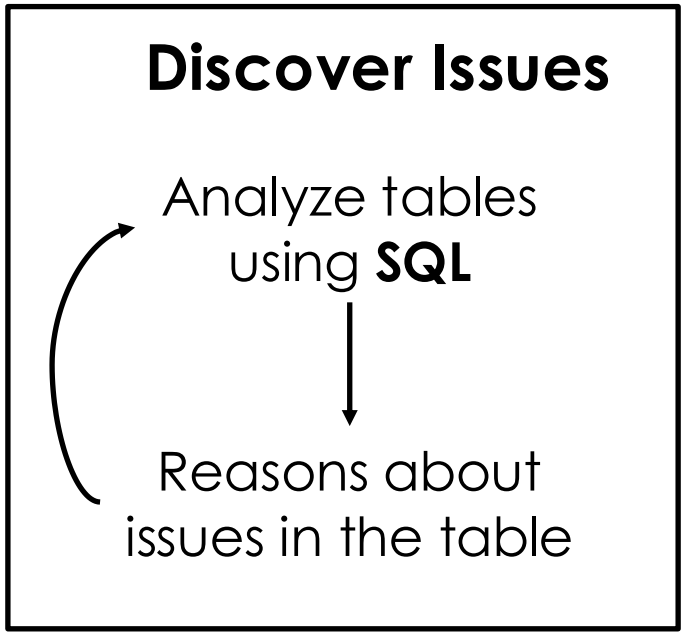
- **Metrics over time:** Key = entity + time_period

- **Events with content:** Key = entity + timestamp

- **Entity relationships:** Key = source + target

- Canonicalize values for consistent results

Autonomous Database Reconciliation Coding Agent

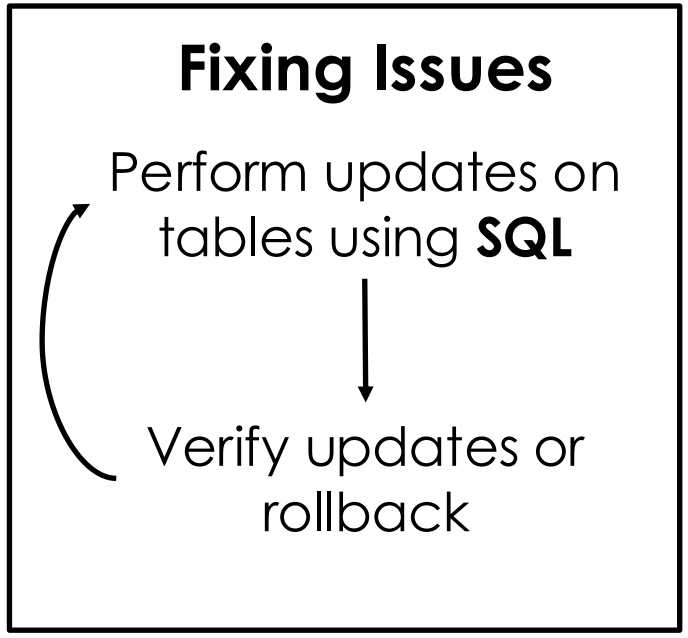


Kinds of errors possible:

- Duplicate
  - **Trigger:** Multiple rows with redundant problems
  - **Purpose:** Remove redundant rows, keeping the best representation
  - **Kinds of fixes:** Select best row
  - If there are different values, nulls its not issue is not duplication
- Requires aggregation:
  - **Trigger:** requires holding multiple values or need to sum, average, etc
  - **Purpose:** Combine multiple valid values into a unified representation
  - **Kinds of fixes:** SUM, ARRAY_AGG, MIN/MAX
- Incomplete information
  - **Trigger:** Information is scattered across rows with incomplete data that can be filled by copying values
  - **Purpose:** Broadcast complete information to fill missing data
  - **Kinds of fixes:** Will copy consistent values without combining or transforming them
  - Returns multiple rows, use when values need copying, not merged
- Conflicting information
  - **Trigger:** Different values exist for a field, and only One is correct
  - **Purpose:** Select the single best value based on evidence, source and exactness
  - **Kinds of fixes:** Evaluate which value to keep and discard others

Default to conflicting information when uncertain about aggregate or resolve conflict

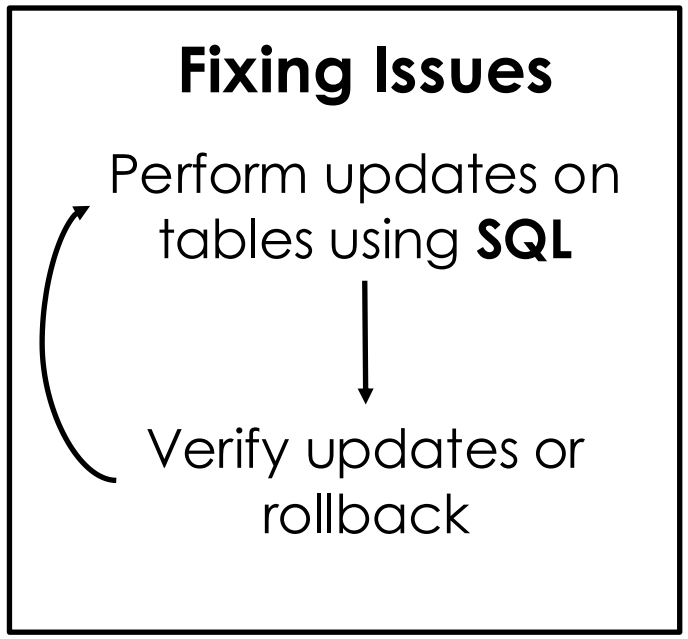
Autonomous Database Reconciliation Coding Agent



- **Instructions**
  - Read the quote and rationale fields to understand what each row represents
  - Compare ALL non-primary-key measured fields (date, amount, description, etc.)
  - Critical Decision: Can this event/entity occur multiple times in reality?
  - Select extracted value over computed
  - Select higher precision value if available

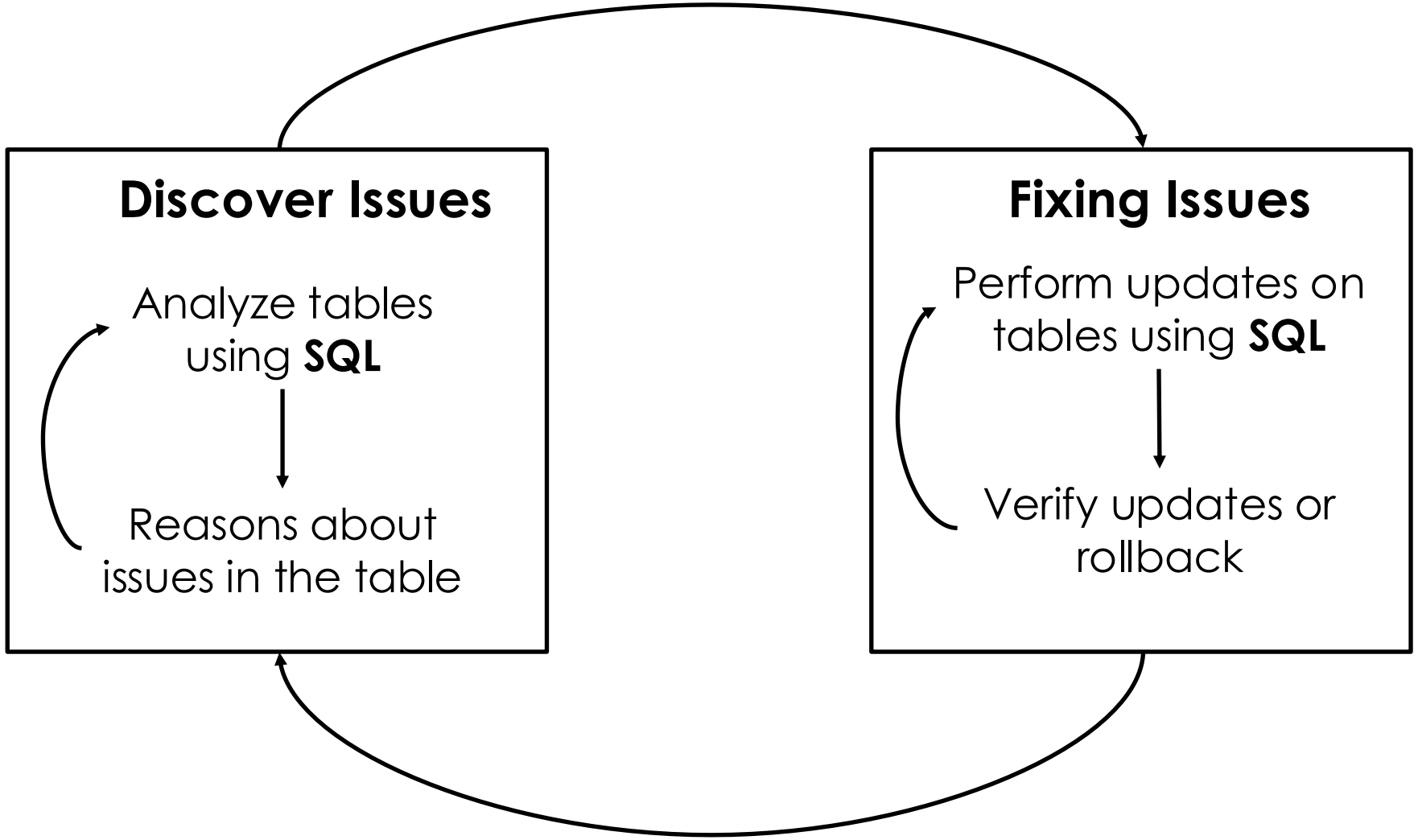
```
SELECT
  * EXCLUDE(col_a, col_a_quote, col_b),
  -- Aggregate multi-valued column across all rows in this PK group
  STRING_AGG(DISTINCT col_a, ' | ') OVER (PARTITION BY pk_col1, pk_col2) AS col_a,
  STRING_AGG(DISTINCT col_a_quote, ' | ') OVER (PARTITION BY pk_col1, pk_col2) AS col_a_quote,
  -- Sum numeric values across rows
  SUM(col_b) OVER (PARTITION BY pk_col1, pk_col2) AS col_b
FROM {{table_name}}
WHERE pk_col1 = '...' AND pk_col2 = '...'
```

Autonomous Database Reconciliation Coding Agent



- We observed that even if LLMs reason correctly for how to fix an issues they still make errors in updates
- Need a verifier that checks if the update on the database matches the intent
  - If not, rollback the changes and give feedback to the LLM.

Autonomous Database Reconciliation Coding Agent



Reasoning for finding issues

There are 8 rows for the same PK ('3M', '2022-12-31'), all in the OperatingMarginMetrics table. The main data columns (operating_income, net_sales, operating_margin) show several repeated values for total company (operating_income=6539, net_sales=34229, operating_margin~19.1), but also several rows for business segments (Safety and Industrial, Transportation and Electronics, Health Care, Consumer) with distinct values. Some rows have calculated margins, some have explicit margins, and some have missing values. The segment rows are not redundant with the total company rows---they represent different aspects. However, among the total company rows, there are clear redundancies (same values, slightly different margin due to rounding/calculation, some with explicit margin, some with calculated). To move toward a single consolidated row, we need to resolve conflicts where the same field has slightly different values (e.g., operating_margin 19.10 vs 19.11), and select the best representative for total company. For the segment rows, aggregation may be needed later, but first, resolving conflicts among total company rows will reduce redundancy and clarify which rows are truly distinct. Therefore, the next step is to resolve conflicts.

Reasoning for fixing the issue

The 8 rows represent both consolidated company-level metrics and business segment breakdowns for 3M for FY2022. There are true conflicts in the measured values for operating_income, net_sales, and operating_margin, but these are not errors---they reflect distinct entities: the consolidated company and its segments (Safety & Industrial, Transportation & Electronics, Health Care, Consumer). The schema is for company-level metrics, but the extraction includes segment-level rows, which are valuable for analysis. However, the merge objective is to deduplicate and resolve conflicts for the company-level row, not to preserve segment rows as separate events.

Upon inspection, two rows (row_id=1 and row_id=10, also row_id=13) provide the consolidated company-level metrics:

- operating_income: 6539
- net_sales: 34229
- operating_margin: 19.1% (explicit in row_id=1, calculated in row_id=10/13)

Row_id=1 is extracted from a table labeled 'Total Company' and provides explicit values for all three metrics, with direct quotes and clear rationale. Row_id=10 and row_id=13 are from the Consolidated Statement of Income and Business Segment Information, also authoritative, but the operating margin is calculated, not quoted. Row_id=1 best matches the field definition alignment, extraction location relevance, and source authority criteria.

Segment rows (row_id=2,4,6,8) are not company-level and should not be merged into the company-level output row. Row_id=0 is a summary/narrative row with less direct metric extraction. Therefore, the best approach is to select row_id=1 as the single authoritative company-level row for FY2022, preserving all original columns

```
SELECT * FROM OperatingMarginMetrics_69ffa681 WHERE row_id = 1
```

SLIDERS design

- Database
- Schematization
- Chunking and Extraction
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- Autonomous database reconciliation coding agent
- **Coding Agent for reconciled database**

Coding Agent for Final Answer

- Autonomous database reconciliation coding agent gives use the reconciled table
- We can now use a coding agent that generates sql to generate answers for us
- Uses SQL, tries to perform group by, get stats, captures provenance from the database
- Generates a final report with caveats (if any)

Coding agent for final answer generation

How do long term borrowing patterns differ across companies and sectors?

row_id	page_number	document_name	company_name	period_end	long_term_borrowings	Sector
4	5	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO., INC.	2025-11-29	214.095	Industrials
12	10	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC.	2025-11-30	1368.738	Financials
13	0	CRACKER_BARREL_OLD_COUNTRY_STORE_INC	CRACKER BARREL OLD COUNTRY STORE, INC	2025-10-31	400.897	Consumer Discretionary
18	0	BLACKBERRY_Ltd	BLACKBERRY Ltd	2025-11-30	196.2	Information Technology
22	2	WINNEBAGO_INDUSTRIES_INC	Winnebago Industries Inc.	2025-11-29	550.0	Consumer Discretionary
27	4	HOOKER_FURNISHINGS_Corp	HOOKER FURNISHINGS Corp	2025-11-02	4.1	Consumer Discretionary
43	3	RESOURCES_CONNECTION_INC.	RESOURCES CONNECTION, INC.	2025-11-29	0.0	Industrials

• • • *108 more rows*

Step 1: before comparing sectors, understand how skewed borrowings are. Bucket companies by size, keep only High or Very High confidence rows, and sum each bucket to see where the debt sits

```
SELECT CASE WHEN long_term_borrowings = 0    THEN '0 (no debt)'
            WHEN long_term_borrowings < 10   THEN '<10M'
            WHEN long_term_borrowings < 100  THEN '10M-100M'
            WHEN long_term_borrowings < 1000 THEN '100M-1B'
            WHEN long_term_borrowings < 5000 THEN '1B-5B'
            ELSE '>5B' END AS bucket,
       COUNT(DISTINCT company_name) AS companies,
       ROUND(SUM(long_term_borrowings), 0) AS total_usd_m
FROM CompanyLongTermBorrowings
WHERE long_term_borrowings IS NOT NULL
  AND long_term_borrowings_confidence IN ('High', 'Very High')
GROUP BY bucket
ORDER BY MIN(long_term_borrowings);
```

bucket	companies	total_usd_m
0 (no debt)	14	0
<10M	31	114
10M-100M	10	489
100M-1B	30	11688
1B-5B	24	67973

Step 2: the top bucket holds half the total with nine companies. Quantify how concentrated debt is before drawing sector conclusions.

```
WITH per_company AS (
  SELECT company_name, MAX(long_term_borrowings) AS v
  FROM CompanyLongTermBorrowings
  WHERE long_term_borrowings IS NOT NULL
  GROUP BY company_name
), ranked AS (
  SELECT *, ROW_NUMBER() OVER (ORDER BY v DESC) AS rk FROM per_company
)
SELECT ROUND(100.0 * SUM(CASE WHEN rk <= 5  THEN v END) / SUM(v), 1) AS top5_share_pct,
       ROUND(100.0 * SUM(CASE WHEN rk <= 10 THEN v END) / SUM(v), 1) AS top10_share_pct,
       COUNT(*) AS companies_with_value
FROM ranked;
```

top5_share_pct	top10_share_pct	companies_with_value
39.1	59.4	109

Step 5: retail has the widest range and the single largest borrower. List its companies to see whether the pattern is uniform or driven by one name, and check for repeated periods.

```
SELECT company_name, period_end, long_term_borrowings
FROM CompanyLongTermBorrowings
WHERE sector = 'Retail' AND long_term_borrowings IS NOT NULL
ORDER BY long_term_borrowings DESC
LIMIT 50;
```

company_name	period_end	long_term_borrowings
CARMAX INC	2025-11-30	17353.935
AUTOZONE INC	2025-11-22	8623.112
Macy's, Inc.	2025-11-01	7243.000
COSTCO WHOLESALE CORP /NEW	2025-11-23	5666.000
GameStop Corp.	2025-11-01	4200.000
Macy's, Inc.	2025-02-01	2779.000
Macy's, Inc.	2024-11-02	2779.000

How do long term borrowing patterns differ across companies and sectors?

Long-term borrowings across the 100 filings are extremely skewed. Of 100 companies with a reported figure, the median is about 123 million USD while the mean is about 1.2 billion, and the five largest borrowers account for 39% of all reported debt, the top ten for 59%. Nine companies above 5 billion USD hold roughly 89.6 billion USD, more than the other hundred combined.

Capital-intensive and consumer-financing sectors carry the most debt. Semiconductors shows the highest median at about 11.8 billion USD, driven by Micron at 11.76 billion and Marvell at 4.5 billion. Retail has the widest spread: CarMax leads the entire dataset at 17.35 billion USD, reflecting its auto-finance receivables funding, followed by AutoZone at 8.6 billion, Macy's at 7.2 billion, Costco at 5.7 billion, and GameStop at 4.2 billion. Software is bimodal: Salesforce reports 8.5 billion and Accenture 5.0 billion, while Okta reports none.

At the other end, 14 companies report zero long-term borrowings and another 31 report under 10 million USD. These are predominantly shell or acquisition companies, early-stage technology firms, and small restaurant operators, which fund themselves through equity rather than term debt.

Two caveats apply. Micron appears under two name spellings with nearly identical values, so sector totals should treat it as one company. Macy's contributes figures for three different period ends, so the 2.78 billion rows reflect earlier comparative periods rather than current borrowings. In addition, 92 of the 233 reconciled rows have no borrowings figure, so companies that did not state a long-term debt line are absent from these comparisons.

Also tell you what might have gone wrong

Ask follow up questions

row_id	page_number	document_name	company_name	period_end	long_term_borrowings	Sector
4	5	MSC_INDUSTRIAL_DIRECT_CO_INC	MSC INDUSTRIAL DIRECT CO., INC.	2025-11-29	214.095	Industrials
12	10	FACTSET_RESEARCH_SYSTEMS_INC	FACTSET RESEARCH SYSTEMS INC.	2025-11-30	1368.738	Financials
13	0	CRACKER_BARREL_OLD_COUNTRY_STORE_INC	CRACKER BARREL OLD COUNTRY STORE, INC	2025-10-31	400.897	Consumer Discretionary
18	0	BLACKBERRY_Ltd	BLACKBERRY Ltd	2025-11-30	196.2	Information Technology
22	2	WINNEBAGO_INDUSTRIES_INC	Winnebago Industries Inc.	2025-11-29	550.0	Consumer Discretionary
27	4	HOOKER_FURNISHINGS_Corp	HOOKER FURNISHINGS Corp	2025-11-02	4.1	Consumer Discretionary
43	3	RESOURCES_CONNECTION_INC.	RESOURCES CONNECTION, INC.	2025-11-29	0.0	Industrials

••• *108 more rows*

- Once we have extracted these values, we can ask follow up questions like the company with minimum debt, with maximum debt, etc.
- We can create the database once and then reuse it with amortized cost

EVALUATION

Lack of benchmarks!

- Real-world applications of systematic reviews often use documents which far exceed the context windows
- No benchmark exists which requires all
  - Extraction from several documents
  - Extract large amount of data
  - Requires comparing information

New Benchmarks

(3.9M tokens)

WikiCeleb100

- 100 Wikipedia pages on recently trending celebrities
- Research Question: Who is the youngest celebrity?

(36M tokens)

FinQ100

- 100 10-Q documents from 2025 Nov – 2026 Jan
- Research Question: Which company has the lowest long-term debt?

Manually annotated reconciled tables
Manually verified final answer
Additional follow-up questions for each domain

Evaluation on existing benchmarks

- Doesn't fit in the context:
  - Browsecomp+: Given several documents answer multi-hop questions
- Fits in the context:
  - FinanceBench: single document
  - Loong: multiple document
  - Oolong: Synthetically created text

Evaluation Results

Table 1: Performance comparison on BrowseComp+ (BC+), Wiki-Celeb100 (WC), and FinQ100 (FQ). $$ means cost prohibitive.

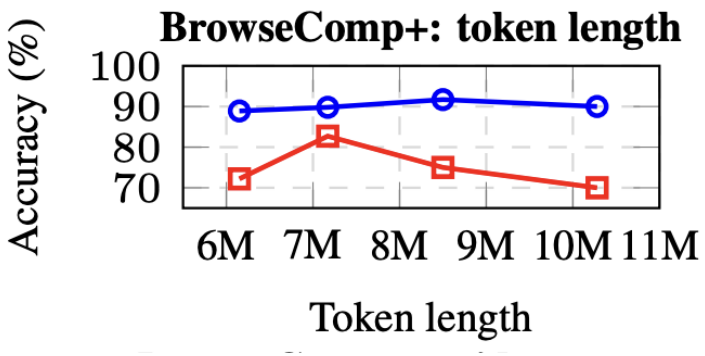
Method	Worker / Reasoning LM	BC+	WC	FQ
RAG	– / GPT-4.1	20.67	31.41	5.00
LongRAG	– / GPT-4.1	28.67	43.20	28.87
GraphRAG	– / GPT-4.1	$$	48.59	08.00[a]
GraphRAG	– / Qwen	$$	48.59	07.00
DocETL	GPT-4.1 / GPT-4.1	70.00	54.26	16.00[a]
RLM	GPT-5-mini / GPT-5	80.00	59.80	28.00
SLIDERS	GPT-4.1-mini / GPT-4.1	**90.00**	**78.91**	55.22
SLIDERS	Qwen3.5 122B / Qwen3.5 122B	–	76.92	**60.18**

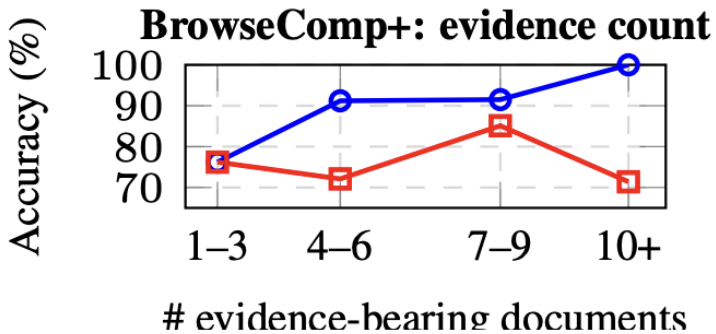
[a] on 10 documents. For the same documents, SLIDERS gets a score of 64.00

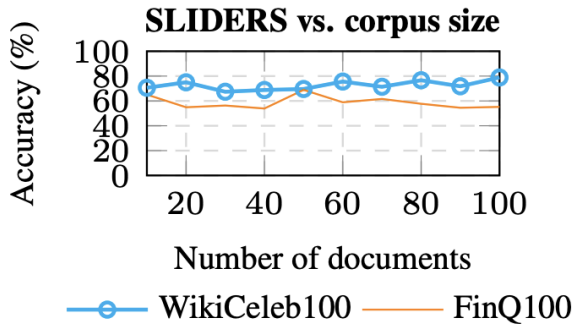
- RAG fails on questions that requires high evidence recall
- Chunking method are better than RAG
- SLIDERS outperforms even with a weaker model
- SLIDERS can reuse its evidence table

Analysis

- SLIDERS mains performance with varying input token length on BrowseComp+
- Even as number of evidence increases, SLIDERS still holds up across BrowseComp+, WikiCeleb100 and FinQ100







SLIDERS RLM

Benchmarks that fit in the context window

Table 3: Performance comparison across FinanceBench (FB), Loong, Oolong, and Average

Method	Worker / Reasoning LM	FB	Loong	Oolong	Avg
RAG	– / GPT-4.1	62.67	54.35	11.32	42.78
LongRAG	– / GPT-4.1	72.00	59.10	22.00	51.03
GraphRAG	– / GPT-4.1	75.33	61.28	22.00	52.87
BaseLM	– / GPT-4.1	82.00	76.74	45.56	68.10
BaseLM	– / Qwen3.5 122B	84.67	74.78	24.89	61.45
DocETL	GPT-4.1 / GPT-4.1	63.33	75.03	49.00	62.45
Chain of Agent	GPT-4.1-mini / GPT-4.1	71.30	54.46	17.11	47.62
RLM	GPT-4.1-mini / GPT-4.1	72.00	60.13	52.00	61.38
RLM	GPT-5-mini / GPT-5	75.33	72.64	51.42	66.46
SLIDERS	GPT-4.1-mini / GPT-4.1	**89.67**	**78.57**	64.67	**77.64**
SLIDERS	Qwen3.5 122B / Qwen3.5 122B	82.10	75.70	**68.00**	75.27

- Keyword search (RLM) can be worse than giving the LLM the whole document
- SLIDERS can even help when the input fits in the LLM's context

Which stage in SLIDERS helps?

- Direct table: provide the extracted table to the LLM
- No Recon: Do not perform reconciliation using autonomous coding agent
- No chunk extraction: Ask the llm to put information in a database direction (when the input fits)

Variants	BC+	WC	FQ
SLIDERS	**90.00**	**78.91**	**55.22**
Direct table	87.33	–	–
No Recon.	87.33	60.50	35.81

Variant	FB	Loong	Oolong
SLIDERS	**80.00**	84.37	**64.67**
Direct Table	70.00	**84.45**	58.62
No Reconciliation	76.70	82.84	62.42
No chunk extraction	70.00	79.72	40.00

Conclusions

- **Important to analyze across large sets of long documents**
  - Need to scale beyond model context limit
- **Precision degrades with document length in the context**
- Approaches
  - **Training: for precision – not successful**
  - **Chunking: for precision and scalability**
  - SLIDERS:
    - **Leverages SUQL: text → databases**
    - **Use Autonomous Coding Agent for reconciliation**
    - **Semantic chunking, contextualized extraction**
- Long-context benchmarks: SOTA (despite annotation errors)
- **Scale beyond long-context benchmarks**